

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 14A.1

THE BRØNSTED-LOWRY THEORY

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	14: Acid-Base Equilibria
SUB-TOPIC:	14A.1: The Brønsted-Lowry Theory
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	3 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 14A.1

Q1. State the Brønsted–Lowry definition of an acid.

[1 Marks]

WCH14/01/Jan23/Q3(a)

Q2. Identify the conjugate acid–base pairs in the following reaction: $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

[2 Marks]

WCH14/01/Oct22/Q11(a)

SECTION B: WORKED MARK SCHEME — 14A.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: State the Brønsted-Lowry definition of an acid. A proton (H^+) donor.	[1]
Q2	Q2: Identify the conjugate acid-base pairs in the following reaction: $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ NH_3 is the base and NH_4^+ is its conjugate acid. (1) H_2O is the acid and OH^- is its conjugate base. (1)	[2]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 14A.1**FAQ #1: What is the most common mistake when identifying conjugate pairs?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What is the most common mistake when identifying conjugate pairs?

MODEL ANSWER / EXAMINER EXPECTATION:

Students often pair up the wrong species. A conjugate pair always differs by exactly one H^+ ion.